

# Ocean Acidification Lab





# Ocean Acidification

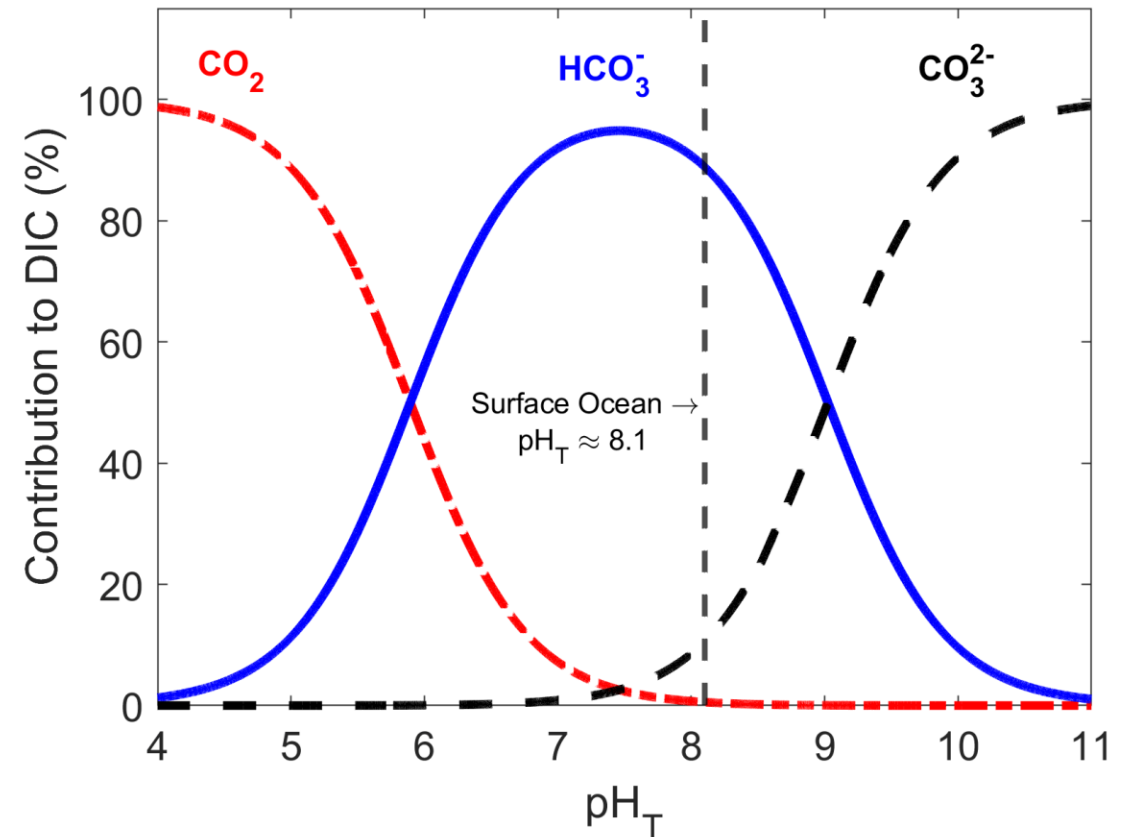
Carbon dioxide combines with seawater to form carbonic acid.



Carbonic acid is unstable and dissociates into bicarbonate

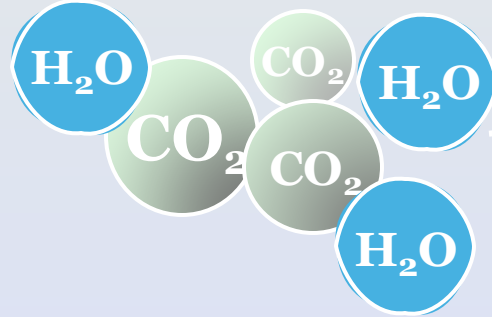


Bicarbonate can further dissociate into carbonate.

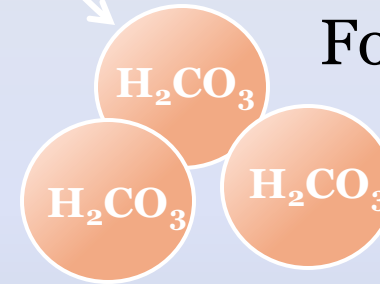


# Ocean Acidification (OA)

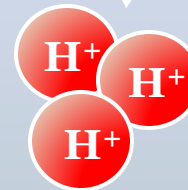
**Surface Ocean**



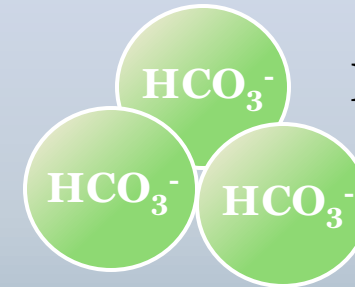
Anthropogenic carbon dioxide ( $\text{CO}_2$ ) dissolves in seawater



Forms carbonic acid



pH lowers ( $[\text{H}^+]$  inc.)

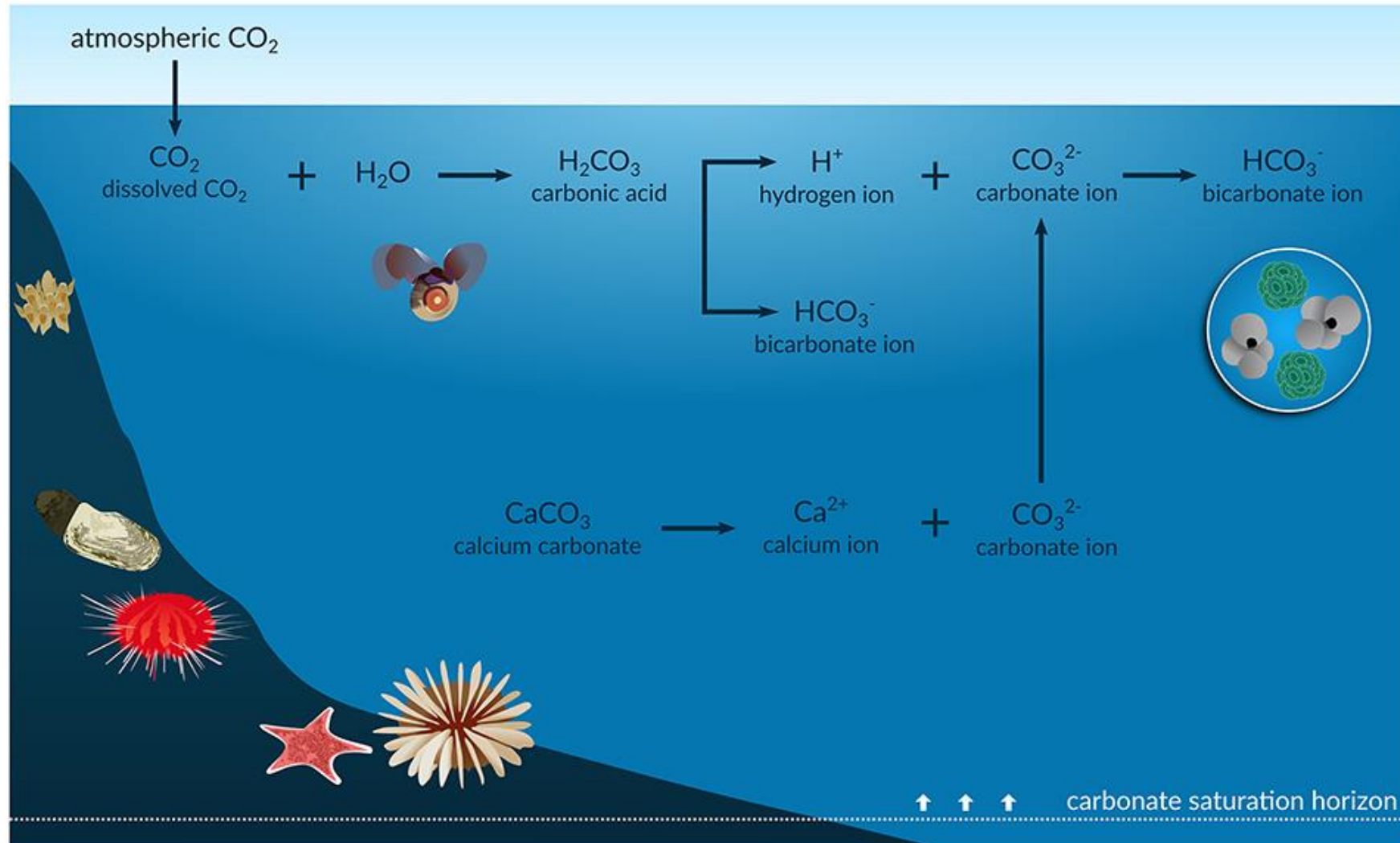


Bicarbonate increases

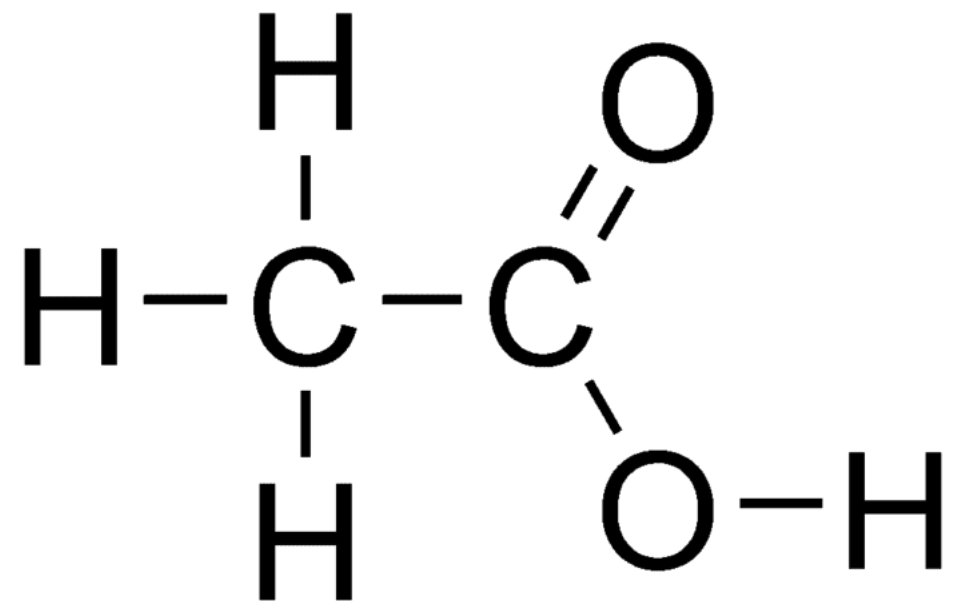


Carbonate depression

# Calcium carbonate - $\text{CaCO}_3$



Vinegar - CH<sub>3</sub>COOH





# Study Area

- Kāne'ohe Bay, Hawai'i
  - Hawai'i Institute of Marine Biology





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*Pocillopora acuta*

# Study Species



*Montipora capitata*





*Pocillopora acuta*



*Montipora capitata*

# What you will be measuring today!

- Linear extension
  - How well are corals growing outward
  - Rapid calcification





# Research Question - Corals

1. How does Ocean Acidification impact the growth of 2 Hawaiian corals, *Montipora capitata* (MC) and *Pocillopora acuta* (PA)?

# Hypothesis - Corals

1. \_\_\_\_\_ will grow less under Ocean Acidification  
than the Control

Or

\_\_\_\_\_ will grow more than \_\_\_\_\_



# Methods - Corals

1. Record species and treatment on your data sheet
2. Measure each coral 3 times and take an average
3. Repeat for all 4 corals
4. Give averages to Alex to add to class datasheet



# What you will be measuring today!

- Dissolution
  - How much the calcium carbonate skeleton dissolves under acidic conditions (low pH)



# Research Question - Oysters

1. How does Ocean Acidification impact oyster shell dissolution?

# Hypothesis - Oysters

Oysters will dissolve \_\_\_\_\_x faster in vinegar (low pH) than in water



# Methods - Oysters

1. Label water and vinegar cups with names
2. Measure the pH of the water and vinegar with the test strips
3. Measure the pH of the water and vinegar with the pH meter
4. Measure both oyster weights and record the weight and treatment on your data sheet
5. Place oysters in water and vinegar cups
6. Record final weights on Thursday
7. Calculate change in weight